

Xiaotian Su

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Education

- **Swiss Federal Institute of Technology Lausanne (EPFL)** **Lausanne, Switzerland**
MSc in Data Science *Sept 2021 - March 2024*
 - Committee Member of [Lausanne Hackathon](#) and [CSSA Lausanne](#).
 - Thesis: Evaluating Dialogue Tutors: Towards Improving the Pedagogical Ability of Language Models.
- **Eötvös Loránd University (ELTE)** **Budapest, Hungary**
BSc in Computer Science, Grade: 4.95 / 5.00 *Sept 2018 - July 2021*
 - Thesis: Building a Tree-LSTM model for Program Synthesis With a Stack-Based Concatenative Language.

Selected Publications

- Unraveling Downstream Gender Bias from Large Language Models: A Study on AI Educational Writing Assistance.
[Xiaotian Su*](#), Thimo Wambsganss*, Vinitra Swamy, Seyed Parsa Neshaei, Roman Rietsche, and Tanja Käser. (* Equal Contribution)
In Findings of the Association for Computational Linguistics: EMNLP 2023
- Reviewer: AI-Generated Instructions For Peer Review Writing.
[Xiaotian Su](#), Thimo Wambsganss, Roman Rietsche, Seyed Parsa Neshaei, and Tanja Käser.
In Proceedings 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023). Association for Computational Linguistics
- Enhancing Peer Review with AI-Powered Suggestion Generation Assistance.
Seyed Parsa Neshaei, Roman Rietsche, [Xiaotian Su](#), Thimo Wambsganss.
In Proceedings of the 29th international conference on intelligent user interfaces. 2024.

Research Experience

- **Autoencoder-based transferability for student success prediction** **Lausanne, Switzerland**
Machine Learning for Education Laboratory (ML4ED), Prof. Tanja Käser *Feb 2022 - June 2022*
 - Conducted time series analysis on students' raw clickstream data to gain insights into their behavior.
 - Developed an encoder-decoder architecture to generate latent feature spaces for students' log data on MOOC.
 - The resulting latent representation achieved a prediction accuracy of over 90% for students' early success.
- **Unianalytics for Master's students admission** **Lausanne, Switzerland**
Computer-Human Interaction in Learning and Instruction Laboratory (CHILI)
Prof. Pierre Dillenbourg *June 2022 - Oct 2022*
 - Performed data cleaning and feature engineering on students' academic records to prepare them for analysis.
 - Developed ML pipelines to predict students' performance and analyze the fairness of models and human decisions.
 - Created a user interface to present students' application materials and visualize prediction results to facilitate data-driven decision-making for the admission committee.
- **Evaluating dialogue tutors (Master thesis)** **Zürich, Switzerland**
Language, Reasoning and Education Laboratory (NLPED)
Prof. Mrinmaya Sachan *Oct 2023 - March 2024*
 - Define the utterance reframing task to give more opportunities to learn using existing learning science work.
 - Annotate existing teacher utterances in dialogue tutoring datasets such as CIMA using our proposed definition.
 - Fine-tune models like T5 or BART on the reframing task and perform a human evaluation for quality check.

Group Projects

- **Robust journey planner on Swiss multimodal transportation network** **Lausanne, Switzerland**
Dependable Systems Laboratory, Prof. George Candea *April 2022 - June 2022*
 - Curating and wrangling large-scale, real-world transportation data from Swiss Federal Railways.
 - Real-time Window-based big data processing using Apache Spark Streaming.
 - Developed a route planner utilizing Dijkstra's algorithm to find the fastest path between two stops.
- **Analysed news quotations to reveal women's underrepresentation** **Lausanne, Switzerland**
Data Science Laboratory (DLAB), Prof. Robert West *Feb 2022 - June 2022*
 - Extracted and wrangled data from [Quotebank](#) (a vast collection of 178 million quotations).
 - Processed and analyzed quotes with NLTK and BERT package, withdrew insights to construct a data story.
 - Designed and implemented a [website](#) to present data story in an interactive and visually appealing way.

Work Experience

- **Kudelski Internet of Things** **Lausanne, Switzerland**
Advanced research team, Data science Intern *Feb 2023 - Aug 2023*
 - Product data management and feature engineering on one terabyte data with more than 200 properties.
 - Machine learning for anomaly detection with Internet of Things (IoT) time series data.**Technical Skills:** Amazon Web Services (AWS), AWS SageMaker, Amazon S3, PySpark.
- **Eötvös Loránd University (ELTE)** **Budapest, Hungary**
Department of Programming Languages and Compilers, Teaching assistant *Sept 2020 - July 2021*
 - Assisted the lecturer to carry out teaching and research work, established and improved the teaching system.
 - Prepared and supervised exercise sessions, and organized curriculum for a class with more than 100 students.
- **Ericsson Telecommunications** **Budapest, Hungary**
MINI-LINK department, Software Developer *Jan 2020 - Aug 2020*
 - Formulated test plans, designed and wrote test scripts according to product requirements.
 - Collaborated with the team to optimize test automation and improve its efficiency.

Awards

Carnegie Mellon University LearnLab Summer School Scholarship (EDM track)	<i>July 2023</i>
2nd ACM Digital Humanism Summer School Scholarship	<i>July 2023</i>
EIT Digital Summer School Scholarship on Data Science for Financial Problems	<i>July 2021</i>
Outstanding student of the faculty (ELTE)	<i>July 2021</i>
EFOP research scholarship (ELTE)	<i>2019 - 2021</i>
Stipendium Hungarian Scholarship (ELTE)	<i>2018 - 2021</i>

Academic Service

- 2024 Help to organize IUI'24 Speech Design Workshop
- 2023-2024 Reviewers of ACM Conference on Human Factors in Computing Systems (CHI)
- 2023-2024 Reviewers of European Conference on Information Systems (ECIS)

Languages

English: Professional **Mandarin:** Native **French, Hungarian:** Beginner